



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session 2026-2027

Vision: Dream of where you want.	Mission: Means to achieve Vision
---	---

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	Preparation	P: Preparation	Pep-CL abbreviation pronounce as Pep-si-IL easy to recall
PEO2	Core Competence	E: Environment (Learning Environment)	
PEO3	Breadth	P: Professionalism	
PEO4	Professionalism	C: Core Competence	
PEO5	Learning Environment	L: Breadth (Learning in diverse areas)	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:

Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

“I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life.” to contribute to the development of cutting-edge technologies and Research.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date

(Signature and Date in Handwritten)



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Session	2026-27	Course Name	Web Technology Lab
Semester	3	Course Code	23CT1301
Roll No	13	Name of Student	Juhi Parwani

Practical Number	1(A)
Course Outcome	1. Understand various internet technologies. 2. Design the web pages using HTML and CSS. 3. Implement the XML technology to store the data. 4. Develop the interactive web pages using JavaScript.
Aim	To develop a Weather Application using JavaScript Fetch API and understand asynchronous programming using Promises, async/await, JSON and try...catch.
Problem Definition	Develop a weather application that accepts a city name from the user, finds its geographical coordinates using a Geocoding API, requests current weather information using the coordinates, and displays the temperature, humidity, wind speed and weather code on the webpage. The application should also handle invalid input, unavailable cities and network/API errors.
Theory (100 words)	Asynchronous programming allows JavaScript to perform time-consuming operations such as API requests without freezing the webpage. An API (Application Programming Interface) provides a way for applications to communicate with external services. In this practical, the fetch() function is used to send network requests and it returns a Promise . The async keyword defines an asynchronous function, while await waits for the Promise to complete. APIs commonly return data in JSON (JavaScript Object Notation) format, which can be converted into JavaScript objects using response.json() . Error handling is performed using try...catch . The received weather data is displayed on the webpage using DOM manipulation and innerHTML .



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

Procedure and Execution

(100 Words)

Step for Implementation:

Open VS Code and create a folder named **weather-app**.
Create three files: **index.html**, **style.css** and **script.js**.
Create an input box for entering the city name.
Create a Search button and weather result container.
Use JavaScript to read the city entered by the user.
Send a request to the Geocoding API using **fetch()**.
Extract latitude and longitude from the API response.
Send another request to the Weather API.
Extract temperature, humidity, wind speed and weather code.
Display the information on the webpage.
Use **try...catch** to handle errors

Code:

```
<!DOCTYPE html>
<html>
<head>
  <title>Weather App</title>
  <link rel="stylesheet" href="style.css">
</head>

<body>

  <h1>Weather App</h1>

  <input type="text" id="cityInput" placeholder="Enter
city name">

  <button id="searchButton">Search</button>

  <div id="weatherResult"></div>

  <script src="script.js"></script>

</body>
</html>
let cityInput = document.getElementById("cityInput");
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

```
let searchButton =
document.getElementById("searchButton");
let weatherResult =
document.getElementById("weatherResult");

searchButton.addEventListener("click", function() {

    let city = cityInput.value.trim();

    if (city == "") {
        weatherResult.innerHTML = "<p>Please enter a
city.</p>";
        return;
    }

    getWeather(city);
});

async function getWeather(city) {

    weatherResult.innerHTML = "<p>Loading...</p>";

    try {

        // Geocoding API
        let geoUrl =

"https://geocoding-api.open-meteo.com/v1/search?name="
        + encodeURIComponent(city)
        + "&count=1";

        let geoResponse = await fetch(geoUrl);
        let geoData = await geoResponse.json();

        if (!geoData.results) {
            weatherResult.innerHTML = "<p>City not
found.</p>";
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

```
        return;
    }

    let location = geoData.results[0];

    let latitude = location.latitude;
    let longitude = location.longitude;

    // Weather API
    let weatherUrl =
        "https://api.open-meteo.com/v1/forecast"
        + "?latitude=" + latitude
        + "&longitude=" + longitude
        +
        "&current=temperature_2m,relative_humidity_2m,"
        + "weather_code,wind_speed_10m";

    let weatherResponse = await fetch(weatherUrl);
    let weatherData = await weatherResponse.json();

    let current = weatherData.current;

    let temperature = current.temperature_2m;
    let humidity = current.relative_humidity_2m;
    let windSpeed = current.wind_speed_10m;
    let weatherCode = current.weather_code;

    // Display result
    weatherResult.innerHTML =
        "<h2>" + location.name + "</h2>" +
        "<p>Country: " + location.country + "</p>" +
        "<p>Temperature: " + temperature + " °C</p>"
        +
        "<p>Humidity: " + humidity + " %</p>" +
        "<p>Wind Speed: " + windSpeed + " km/h</p>" +
        "<p>Weather Code: " + weatherCode + "</p>";
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

```
}

catch(error) {

    weatherResult.innerHTML =

        "<p>Unable to fetch weather data.</p>";

    console.log(error);

}

body {

    font-family: Arial;
    text-align: center;
    margin-top: 50px;

}

input {

    padding: 10px;
    width: 250px;

}

button {

    padding: 10px 20px;
    cursor: pointer;

}

#weatherResult {

    margin-top: 30px;

}
```



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



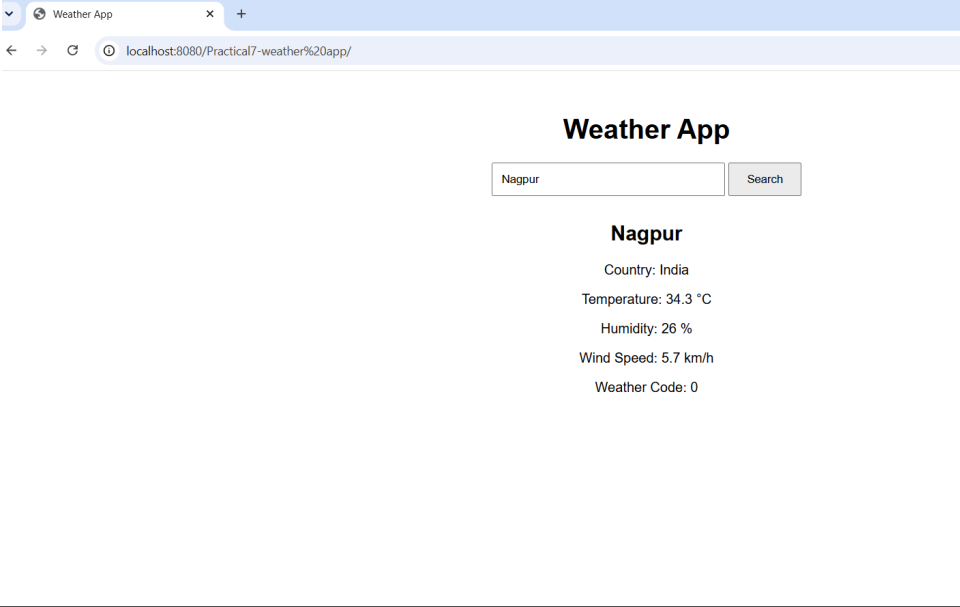
Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

	Output: 
Output Analysis	The application successfully accepts a city name and retrieves weather information through external APIs. The Geocoding API provides latitude and longitude, which are then used by the Weather API to obtain current weather information. The data is converted from JSON into JavaScript objects and displayed dynamically using DOM manipulation.
Link of student Github profile where lab assignment has been uploaded	https://github.com/juhiparwa-1507/Web-Technology-Lab.git
Conclusion	Thus, the Weather Application was successfully developed using HTML, CSS and JavaScript . The practical demonstrates asynchronous JavaScript concepts including API, Promise, fetch(), async/await, JSON, DOM manipulation and try...catch error handling .
Plag Report (Similarity index < 12%)	Plagiarism Report: To be attached.



Nagar Yuwak Shikshan Sanstha's

Yeshwantrao Chavan College of Engineering

(An Autonomous Institution affiliated to Rashtrasant Tukadoji Maharaj Nagpur University)

Hingna Road, Wanadongri, Nagpur - 441 110

NAAC A++

Ph.: 07104-237919, 234623, 329249, 329250 Fax: 07104-232376, Website: www.ycce.edu



Department of Computer Technology

Vision Statement of the Department.

To be a preferred hub for nurturing computer engineering aspirants through innovative, ethical, value-based technical education, impactful research, and industry collaboration, to become successful professionals contributing for the betterment of society.

Mission Statement of the Department.

To nurture computer engineering professionals by offering Dynamic learning environment that encourages problem-solving using emerging technology, National Education Policy aligned evidence-based technical education, fostering innovation, research, and industry-institute collaboration & Holistic, ethical and sustainable development for lifelong success.

	Plagiarism Detection Report by SmallSEOTOOLS			
				
	Scan details			
	Total Words 577	Total Characters 4946	Plagiarized Sentences 0	Unique Sentences 20 (100%)
Date	04-10-26			